

Orthogonal Vectors and Subspaces (正交向量和正交子空间)

Lecture 13

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Orthogonal Vectors and Subspaces

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Orthogonality

- A basis is a set of independent vectors that span a space.
- Geometrically, it is a set of coordinate axes. In choosing a basis, we tend to choose an orthogonal basis.
- The idea of an orthogonal basis is one of the foundations of linear algebra.
- We need a basis to convert geometric constructions into algebraic calculations, and we need an orthogonal basis to make those calculations simple.
- A further specialization makes the basis just about optimal: The vectors should have length 1.

Orthogonality

For an orthonormal basis(orthogonal unit vectors), we will find

- (1) the length $\|x\|$ of a vector.
- (2) the test $x^T y = 0$ for perpendicular vectors; and
- (3) how to create perpendicular vectors from linearly independent vectors.

More than just vectors, subspaces can also be perpendicular.

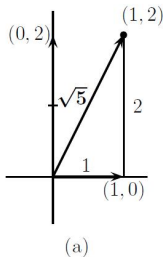
- (a) We will discover, so beautifully and simply that it will be a delight to see, that the fundamental subspaces meet at right angles.
- (b) Those four subspaces are perpendicular in pairs, two in $\mathbb{R}^m$ and two in $\mathbb{R}^n$.
- (c) That will complete the fundamental theorem of linear algebra.

Length of a vector

The first step is to find the length of a vector.

Definition

The length $\|x\|$ in $\mathbb{R}^n$ is the positive square root of $x^T x$.



$$\|x\|^2 = x_1^2 + x_2^2 + x_3^2$$

$$5 = 1^2 + 2^2$$

$$14 = 1^2 + 2^2 + 3^2$$

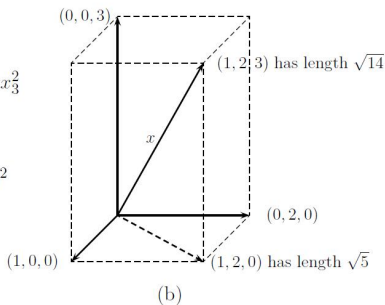


Figure 3.1: The length of vectors (x_1, x_2) and (x_1, x_2, x_3) .

Orthogonal Vectors

How can we decide whether two vectors x and y are perpendicular?

What is the test for orthogonality in Figure 3.2?

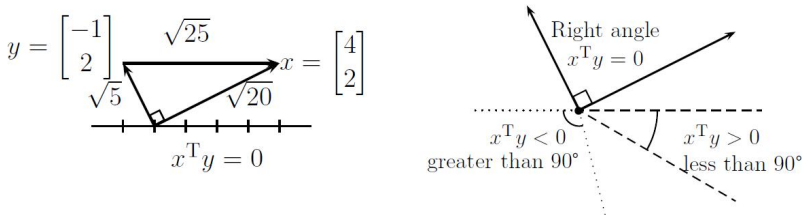


Figure 3.2: A right triangle with $5 + 20 = 25$. Dotted angle 100° , dashed angle 30° .

Inner Product

In the plane spanned by x and y , those vectors are orthogonal provided they form a right triangle. We go back to $a^2 + b^2 = c^2$:

$$\|x\|^2 + \|y\|^2 = \|x - y\|^2.$$

Sides of a right triangle

$$\|x\|^2 + \|y\|^2 = \|x - y\|^2.$$

Inner Product

Applying the length formula, this test for orthogonality in $\mathbb{R}^n$ becomes

$$(x_1^2 + x_2^2 + \cdots + x_n^2) + (y_1^2 + y_2^2 + \cdots + y_n^2) = (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2.$$

The right-hand side has an extra $-2x_i y_i$ from each $(x_i - y_i)^2$:

$$\text{Right-hand side} = (x_1^2 + \cdots + x_n^2) - 2(x_1 y_1 + \cdots + x_n y_n) + (y_1^2 + \cdots + y_n^2)$$

We have a right triangle when that sum of cross-product terms $x_i y_i$ is zero:

$$\text{Orthogonal vectors} \quad x^T y = x_1 y_1 + \cdots + x_n y_n = 0$$

Orthogonal Vectors

This sum is $x^T y = \sum x_i y_i = y^T x$, the row vector x^T times the column vector y :

$$x^T y = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = x_1 y_1 + \cdots + x_n y_n$$

This number is sometimes called the scalar product or dot product, and denoted by (x, y) or $x \cdot y$. We will use the name inner product and keep the notation $x^T y$.

Definition

Definition

The inner product $x^T y$ is zero if and only if x and y are orthogonal vectors. If $x^T y > 0$, their angle is less than 90° . If $x^T y < 0$, their angle is greater than 90° .

Useful Fact

Proposition

If nonzero vectors $v_1, v_2, \dots, v_k$ are mutually orthogonal (every vector is perpendicular to every other), then those vectors are linearly independent.

Proof.

Suppose $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = \mathbf{0}$. Taking the inner product of both sides with v_1 to obtain

$$v_1^T (c_1 v_1 + c_2 v_2 + \dots + c_k v_k) = c_1 v_1^T v_1 = v_1^T \cdot \mathbf{0} = 0.$$

Since v_1 is nonzero, therefore $c_1 = 0$. A similar argument shows all c_i 's are zero. Thus, $v_1, v_2, \dots, v_k$ are linearly independent by definition. □

Examples

1. The coordinate vectors $e_1, e_2, \dots, e_n$ in $\mathbb{R}^n$ are the most important orthogonal vectors. Those are the columns of the identity matrix. They form the simplest basis for $\mathbb{R}^n$, and they are unit vectors—each has length $\|e\| = 1$. They point along the coordinate axes.
2. Orthogonal vectors in $\mathbb{R}^2$. If these axes are rotated, the result is a new **orthonormal basis**: a new system of mutually orthogonal unit vectors. In $\mathbb{R}^2$, we have $\cos^2 \theta + \sin^2 \theta = 1$. Orthonormal vectors in $\mathbb{R}^2$:

$$v_1 = (\cos \theta, \sin \theta) \quad \text{and} \quad v_2 = (-\sin \theta, \cos \theta)$$

Orthogonal Subspaces

- A line can be orthogonal to another line, or it can be orthogonal to a plane, but a plane cannot be orthogonal to a plane.
- We come to the orthogonality of two subspaces.
- Every vector in one subspace must be orthogonal to every vector in the other subspace.

Definition

Two subspaces V and W of the same space $\mathbb{R}^n$ are orthogonal if **every** vector v in V is orthogonal to **every** vector w in W : $v^T w = 0$ for all v and w .

Fundamental Theorem of Orthogonality

- The important orthogonal subspaces don't come by accident, and they come two at a time. In fact orthogonal subspaces are unavoidable: They are the fundamental subspaces!
- The first pair is the nullspace and row space. Those are subspaces of $\mathbb{R}^n$ —the rows have n components and so does the vector x in $Ax = 0$. We have to show, using $Ax = 0$, that the rows of A are orthogonal to the nullspace vector x .

Theorem

The row space is orthogonal to the nullspace (in $\mathbb{R}^n$). The column space is orthogonal to the left nullspace (in $\mathbb{R}^m$).

Let us prove this theorem in two different ways!

Orthogonal complement

It is certainly true that the nullspace is perpendicular to the row space—but it is not the whole truth. $N(A)$ contains **every** vector orthogonal to the row space. The nullspace was formed from all solutions to $Ax = 0$.

Definition

Given a subspace V of $\mathbb{R}^n$, the space of **all** vectors orthogonal to V is called the **orthogonal complement** of V . It is denoted by $V^\perp = "V \text{ perp.}"$

Using this terminology, the nullspace is the orthogonal complement of the row space:

$$N(A) = (C(A^T))^\perp, C(A^T) = (N(A))^\perp.$$

The same reasoning applied to A^T produces the dual result.

Fundamental Theorem of Linear Algebra, Part II

Theorem

*The nullspace is the orthogonal complement of the row space in $\mathbb{R}^n$.
The left nullspace is the orthogonal complement of the column space in $\mathbb{R}^m$.*

Theorem

$Ax = b$ is solvable if and only if $y^T b = 0$ whenever $y^T A = 0$.

Remarks:

- b must be a combination of the columns.
- b must be orthogonal to every vector that is orthogonal to the columns.

The Matrix and the Subspaces

We emphasize that V and W can be orthogonal without being complements.

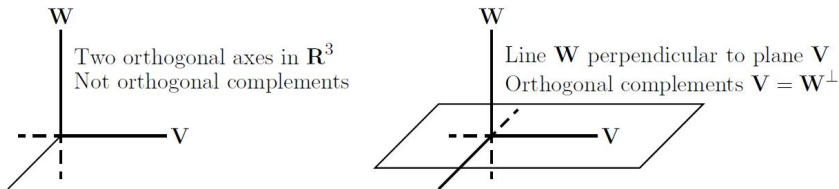


Figure 3.3: Orthogonal complements in $\mathbb{R}^3$: a plane and a line (not two lines).

Remarks:

- Splitting $\mathbb{R}^n$ into orthogonal parts will split every vector into $x = v + w$.
- The vector v is the projection onto the subspace V . The orthogonal component w is the projection of x onto W .

Theorem

Theorem

From the row space to the column space, A is actually invertible. Every vector b in the column space comes from exactly one vector x_r in the row space.

Proof.

Every b in the column space is a combination Ax of the columns. In fact, b is Ax_r , with x_r is in the row space, since the nullspace component gives $Ax_n = 0$. If another vector x'_r in the row space gives $Ax'_r = b$, then $A(x_r - x'_r) = b - b = 0$. This puts $x_r - x'_r$ in the nullspace and the row space, which makes it orthogonal to itself. Therefore it is zero, and $x_r = x'_r$. Exactly one vector in the row space is carried to b . □

Figure 3.4

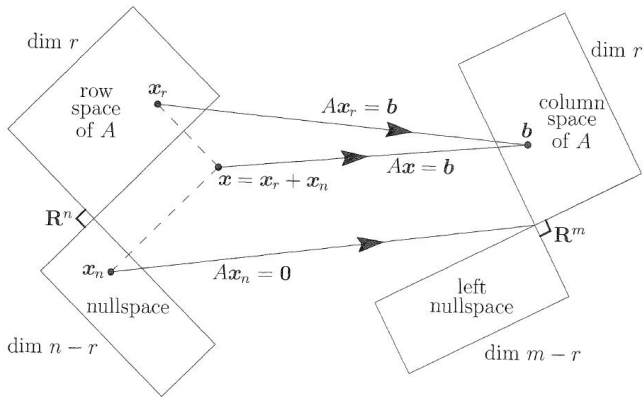


Figure 3.4: The true action $Ax = A(x_{\text{row}} + x_{\text{null}})$ of any m by n matrix.

Proposition

Proposition

Every matrix transforms its row space onto its column space.

Remarks:

- On those r -dimensional spaces A is invertible.
- On its nullspace A is zero.
- A^T goes in the opposite direction, from $\mathbb{R}^m$ to $\mathbb{R}^n$ and from $C(A)$ back to $C(A^T)$.
- When A^{-1} fails to exist, the best substitute is the pseudoinverse A^+ .
- One formula for A^+ depends on the singular value decomposition—for which we first need to know about eigenvalues.

Homework Assignment 13

3.1: 6, 7, 12, 32, 38, 49.